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Purpose of this Overview

This document introduces a conceptual systems architecture for technical review and discussion, outlining the context, scope, boundaries, and operational role of AstroGit-OS empowering observatory control, astronomical operations, and maintenance engineering teams.

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Complete technical drafts are available upon request. **Technical discussions** are welcome.

AstroGit-OS — Overview

GitOps Architecture for State Governance and Operational Lineage of Astronomical Instrumentation

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v0.2 – **Systems Architecture Concept**
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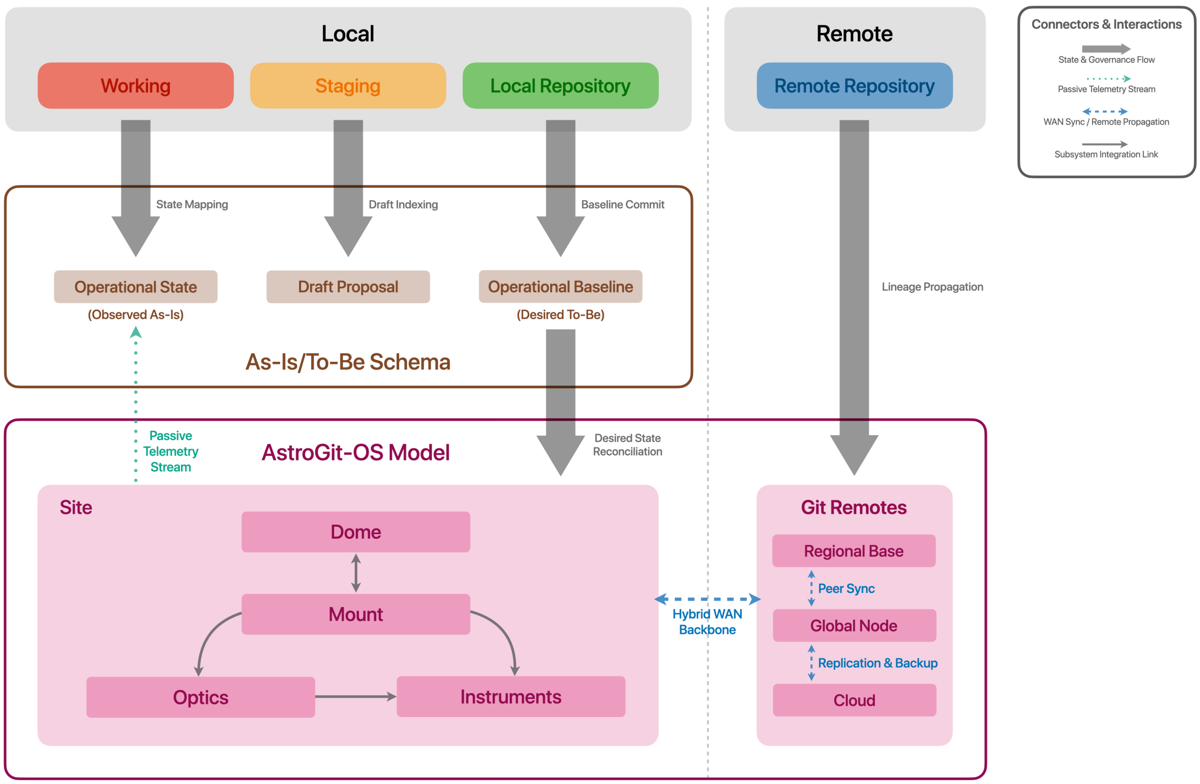
EXECUTIVE SUMMARY

AstroGit-OS is a reference declarative architecture for the configuration, maintenance, and operational lineage of optomechanical instrumentation in next-generation astronomical observatories. Applying GitOps and Infrastructure-as-Code principles, AstroGit-OS dispenses with massive astronomical data and focuses exclusively on the operational metadata required to govern scientific infrastructure.



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The Value of Operational Certainty in Astronomy

Through an operational state reconciliation architecture, AstroGit-OS governs the evolution of infrastructure and instrumentation, detects deviations between observed state and desired configuration, and establishes a verifiable link between observation blocks and the versioned operational state through passive telemetry and non-intrusive state extraction agents. In doing so, it aims to reduce operational uncertainty, minimize downtime, and facilitate the reproduction of technical conditions underlying the scientific use of astronomical instrumentation.

Scope and Boundaries of AstroGit-OS

AstroGit-OS does not capture, store, process, simulate, or transmit astronomical data, nor does it operate telescopes in real time. Its function is to govern a versioned, asynchronous digital representation of the operational state of telescopes and other scientific infrastructure, decoupling scientific activities from engineering operations. **AstroGit-OS records, stores, and audits the evolution of infrastructure configuration through Git as a version-controlled governance mechanism**, keeping the declarative representation of operational state separate from its physical implementation across the instrumentation.

AstroGit-OS Differentiating Principles

- **Scientific Lineage Without Massive Data**
  Associates each astronomical observation block with a SHA-256 identifier of the hardware operational state, enabling verifiable scientific traceability without burdening the repository with massive datasets.

- **Distributed Operational Resilience**
  Supports operational continuity and the recovery of versioned configurations at remote high-altitude observatories through decentralized and synchronized Git topologies.

- **Non-Intrusive Declarative Governance**
  Translates infrastructure maintenance into an auditable workflow through Pull Requests and human approval, operating strictly asynchronously without interfering with real-time telescope operations.

CONTEXTUAL FRAMEWORK AND OPERATIONAL GOVERNANCE

This framework examines the critical challenges facing contemporary optomechanics and the implications of the transition toward the exabyte era, establishing the foundations of AstroGit-OS as a declarative governance model for world-class astronomical instrumentation.

Operational Challenges of Astronomical Optomechanics

Large-scale observatories operate in highly precise optomechanical environments, where thermal, mechanical, optical, and alignment variations can affect the conditions under which scientific data are acquired. The operation of this infrastructure generates information distributed across control systems, databases, maintenance records, and configuration mechanisms, each serving specific operational functions.

This heterogeneity can fragment the historical representation of infrastructure state, making it difficult to reconstruct as a coherent, contextualized, and verifiable record. Addressing this challenge requires more than simply recording events, configurations, or states. **It requires preserving the ability to reproducibly reconstruct the operational state of the infrastructure throughout its lifecycle.** This, in turn, requires linking configuration, observed state, authorized changes, and operational context within a consistent historical representation that remains independent of the original source and can be subsequently verified.

Scale, Complexity, and Automation of Scientific Infrastructure

Modern astronomy is undergoing a profound transformation of scale driven by next-generation projects—such as the Extremely Large Telescope (ELT), the Giant Magellan Telescope (GMT), and the Vera C. Rubin Observatory—operating at the technological frontier. The convergence of high-performance computing, unprecedented optomechanical complexity, and the accelerated growth of scientific data streams poses new challenges for astronomical infrastructure management. As these observatories expand in capacity and complexity, operational infrastructure must evolve alongside them, reducing friction, overhead, and time spent on operational tasks that do not add scientific value. Consequently, automating operational tasks becomes an essential component to ensure that system complexity translates tangibly into reduced operational burden and increased scientific capacity.

Petabytes, Exabytes, and the Reproducibility of Scientific Conditions

Sustaining the distributed digital infrastructure required by astronomy in the coming decades demands a scale shift from today's petabytes to exabytes. This new paradigm—led by ground-based megainfrastructures (ELT, GMT, Rubin) as well as next-generation space missions (Roman, HWO)—creates scenarios where operational continuity and technical reproducibility become essential to support the scientific validity of data based on the technical rigor of astronomical observations.